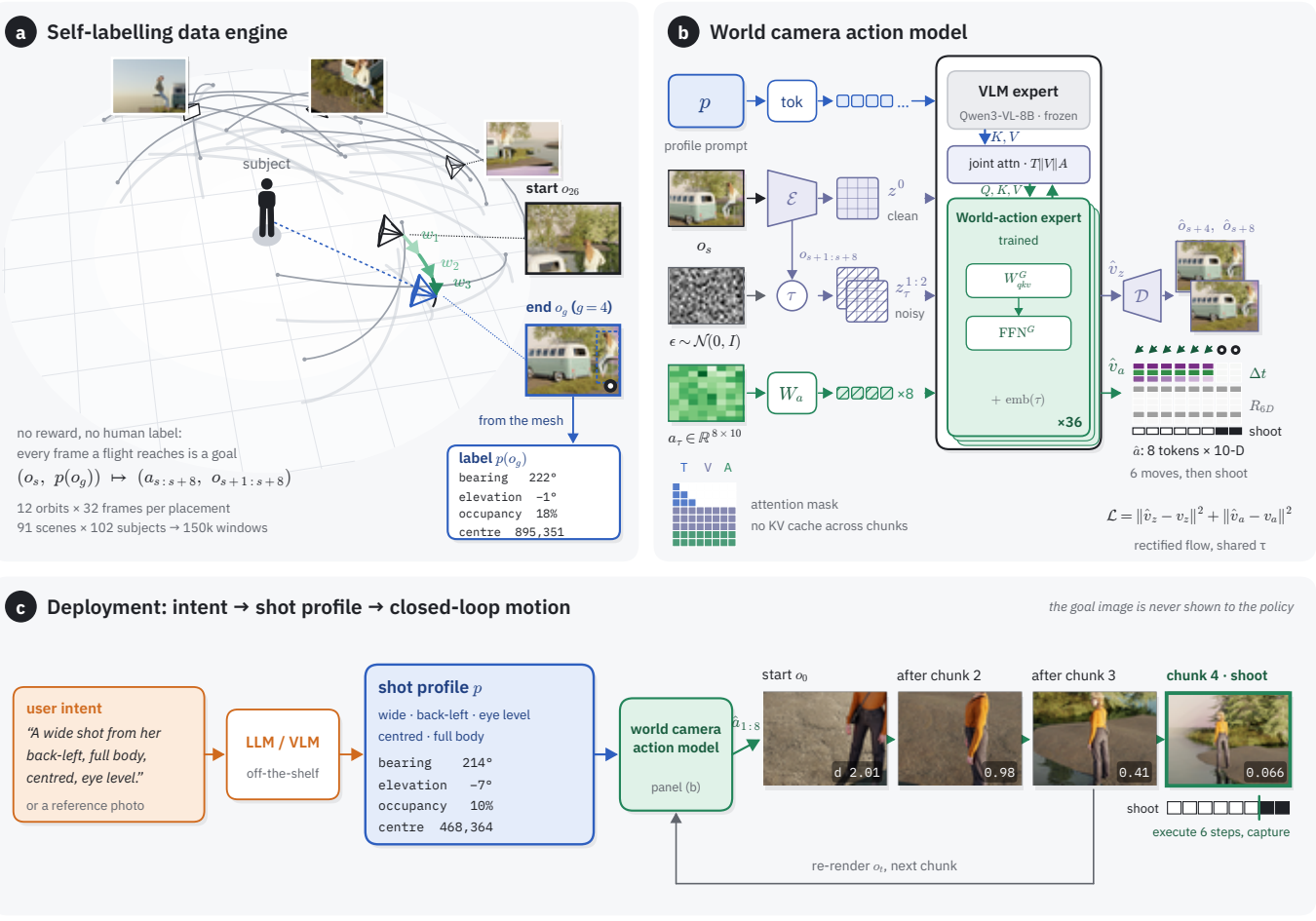

a
Self-labelling data engine
subject
start $o_{26}$
end $o_g$ $(g = 4)$
$w_1$
$w_2$
$w_3$
from the mesh
label $p(o_g)$
bearing 222°
elevation −1°
occupancy 18%
centre 895,351
no reward, no human label:
every frame a flight reaches is a goal
$(o_s,\ p(o_g)) \mapsto (a_{s:s+8},\ o_{s+1:s+8})$
12 orbits × 32 frames per placement
91 scenes × 102 subjects → 150k windows
b
World camera action model
$p$
tok
profile prompt
$o_s$
$\epsilon \sim \mathcal{N}(0, I)$
$a_\tau \in \mathbb{R}^{8\times 10}$
T V A
attention mask
no KV cache across chunks
$\mathcal{E}$
$z^0$
clean
$o_{s+1:s+8}$
$\tau$
$z_\tau^{1:2}$
noisy
$W_a$
×8
VLM expert
Qwen3-VL-8B · frozen
$K, V$
joint attn · $T\|V\|A$
$Q, K, V$
World-action expert
trained
$W_{qkv}^G$
$\mathrm{FFN}^G$
$+\ \mathrm{emb}(\tau)$
×36
$\hat{v}_z$
$\mathcal{D}$
$\hat{o}_{s+4},\ \hat{o}_{s+8}$
$\hat{v}_a$
$\Delta t$
$R_{6D}$
shoot
$\hat{a}$: 8 tokens × 10-D
6 moves, then shoot
$\mathcal{L} = \|\hat{v}_z - v_z\|^2 + \|\hat{v}_a - v_a\|^2$
rectified flow, shared τ
c
Deployment: intent → shot profile → closed-loop motion
the goal image is never shown to the policy
user intent
"A wide shot from her back-left, full body, centred, eye level."
or a reference photo
LLM / VLM
off-the-shelf
shot profile $p$
wide · back-left · eye level
centred · full body
bearing 214°
elevation −7°
occupancy 10%
centre 468,364
world camera action model
panel (b)
$\hat{a}_{1:8}$
start $o_0$
after chunk 2
after chunk 3
chunk 4 · shoot
d 2.01
0.98
0.41
0.066
shoot
execute 6 steps, capture
re-render $o_t$, next chunk